

Exercise 1: Configuring a Basic Spring Application

pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>org.library</groupId>
  <artifactId>Springcore_and_maven</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <maven.compiler.source>26</maven.compiler.source>
    <maven.compiler.target>26</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>
  <dependencies>

    <!-- Spring Core -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>6.2.8</version>
    </dependency>

  </dependencies>

</project>
```

ApplicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">

<bean id="bookRepository"
    class="org.library.repository.BookRepository"/>
<bean id="bookService"
    class="org.library.service.BookService">

    <property name="bookRepository"
        ref="bookRepository"/>

</bean>

</beans>
```

BookRepository.java

```
package org.library.repository;

public class BookRepository {
    public void displayBooks() {
        System.out.println("Fetching books from the repository...");
    }
}
```

BookService.java

```
package org.library.service;

import org.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void showBooks() {
        System.out.println("BookService is calling BookRepository...");
    }
}
```

```
        bookRepository.displayBooks();
    }

}
```

MainApp.java

```
package org.library;

import org.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class MainApp {
    public static void main(String[] args) {

        // Load Spring XML Configuration
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        // Get Bean
        BookService service =
            context.getBean("bookService", BookService.class);

        // Call Method
        service.showBooks();
    }
}
```

